

TOTUS TUUS KEELY

MECHATRONICS ENGINEERING | MECHANICAL DESIGN | AUTOMATION | ROBOTICS

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PROFESSIONAL SUMMARY

Mechatronics Engineering student graduating May 2027 with three internships in industrial equipment design, automated machinery, and manufacturing operations. Experienced with SolidWorks assemblies, sheet metal, weldments, drawings, PDM, BOMs, testing, and design for manufacturability, with additional leadership experience managing a 50-person college news organization.

EDUCATION

Northern Illinois University | B.S. in Mechatronics Engineering Expected May 2027

GPA: 3.822 | Dean's List | DeKalb, IL | Coursework: Robotics and Automation; Dynamic Systems and Control; Fundamentals of Mechatronics; Mechanism Design and Analysis; Signals and Systems; Computer Engineering

TECHNICAL SKILLS

CAD & Documentation: SolidWorks, large assemblies, sheet metal, weldments, drawings, SolidWorks PDM, AutoCAD, Autodesk Inventor, BOM development

Automation & Programming: RoboDK, robot kinematics, coordinate frames, motion planning, collision checking, Python, C++, MATLAB

Embedded & Manufacturing: Raspberry Pi 5, ESP32, Arduino Nano RP2040, DFM, validation testing, CNC routers, mills, lathes, band saws, 3D printing

ENGINEERING EXPERIENCE

Precision Quincy Inc. | Mechanical Engineering Intern South Beloit, IL | May 2026 - Aug. 2026

- Developed the mechanical design of an industrial crossdraft spray booth comprising approximately 100 total parts across 8 assemblies using SolidWorks large assemblies, sheet metal, weldments, and envelope-based modeling.
- Produced manufacturing and assembly drawings, flat patterns, DXFs, cut lists, and multiple BOM types supporting purchasing, cutting, assembly, and complete parts tracking
- Evaluated pressure drop, filter velocity, concrete-anchor strength, and door-hinge strength; selected purchased components through market and cost analysis; referenced NFPA 33, OSHA 1910.107, and applicable laser-safety standards.
- Presented designs to lead engineers and floor managers, incorporated approximately 10 design changes, and completed a 500-sample capacitor-discharge weld gun evaluation across 5 variables with an accompanying SOP and test document.

Pregis IQ | Mechanical Engineering Intern Aurora, IL | May 2025 - Aug. 2025

- Designed approximately 50 parts and 6 SolidWorks assemblies for automated packaging equipment, created drawings for all designed parts and assemblies, and managed design data in SolidWorks PDM.
- Developed a custom test fixture through 4 prototype iterations and procedures demonstrating approximately 40% greater machine efficiency; authored 3 user and technical manuals with specification sheets.

American Ordnance | Operations Engineering Intern Middletown, IA | Jun. 2024 - Aug. 2024

- Reduced rework on one production line from 10% to 5% and built Excel databases and hours-per-unit tracking tools that documented a 30% summer improvement.

LEADERSHIP EXPERIENCE

Northern Star Media | Editor in Chief DeKalb, IL | May 2026 - Present

- Lead approximately 50 staff members with 14 direct reports.
- Run weekly leadership meetings and oversee production of at least eight digital stories daily, three print editions per semester, and a daily five-story newsletter.
- Interviewed approximately 50 applicants for newsroom positions.

SELECTED ENGINEERING PROJECTS

Industrial Robot Pick-and-Place Simulation | Robotics and Automation Spring 2026

- Programmed a 6-axis HIWIN RA605-710-GB robot with a RobotiQ 2F-85 gripper in RoboDK using the Python API for frames, 4x4 transformations, forward and inverse kinematics, motion planning, rotation-matrix validation, and collision checking.

Autonomous Quadruped Navigation Robot | Fundamentals of Mechatronics Fall 2025

- Co-developed Python navigation software for a Raspberry Pi 5 and ESP32-based 12-DOF quadruped, replacing an impractical LiDAR and magnetometer concept with servo-mounted ultrasonic sensing for object avoidance, wall detection, and wall following.

CERTIFICATIONS, AWARDS, TRAINING

FAA Part 107 Remote Pilot Certificate; NVIDIA Fundamentals of Deep Learning; LinkedIn Learning - SOLIDWORKS 2022 Essential Training; Northern Star Most Dedicated Staffer